

SCQ (Single Correct Type) :

- If a_r is the coefficient of x^r in the expansion of $(1 + x + x^2)^n$ ($n \in \mathbb{N}$). Then the value of $(a_1 + 4a_4 + 7a_7 + 10a_{10} + \dots)$ is equal to :
 (A) 3^{n-1} (B) 2^n (C) $\frac{1}{3} \cdot 2^n$ (D) $n \cdot 3^{n-1}$
- All the five digits number in which each successive digit exceeds its predecessor are arranged in the increasing order of their magnitude. The 97th number in the list does not contain the digit
 (A) 4 (B) 5 (C) 7 (D) 8
- Six cards are drawn one by one from a set of unlimited number of cards, each card is marked with numbers $-1, 0$ or 1 . Number of different ways in which they can be drawn if the sum of the numbers shown by them vanishes, is:
 (A) 111 (B) 121 (C) 141 (D) none
- Let $A = \{1, 2, 3, 4\}$ and f and g be randomly chosen (not necessarily distinct) functions defined from set A to A . The probability that the range of f and g are disjoint is given by $\frac{m}{2^n}$, where m and n are positive integers and $\text{hcf}(m, 2^n) = 1$, then
 (A) $m = 453$ (B) $n = 12$ (C) $n = 13$ (D) $m + n = 466$
- Let $x - \frac{1}{x} = (\sqrt{2})i$ where $i = \sqrt{-1}$. Then the value of $x^{2187} - \frac{1}{x^{2187}}$ is :
 (A) $i\sqrt{2}$ (B) $-i\sqrt{2}$ (C) -2 (D) $\frac{i}{\sqrt{2}}$
- From a pack of 52 playing cards; half of the cards are randomly removed without looking at them. From the remaining cards, 3 cards are drawn randomly. The probability that all are king.
 (A) $\frac{1}{(25)(17)(13)}$ (B) $\frac{1}{(25)(15)(13)}$ (C) $\frac{1}{(52)(17)(13)}$ (D) $\frac{1}{(13)(51)(17)}$

MCQ (One or more than one correct) :

- If $a = \sum_{r=0}^{20} {}^{20}C_r$, $b = \sum_{r=0}^9 {}^{20}C_r$, $c = \sum_{r=11}^{20} {}^{20}C_r$, then
 (A) $a = b + c$ (B) $b = 2^{19} - \frac{1}{2} {}^{20}C_{10}$
 (C) $c = 2^{19} + \frac{1}{2} {}^{20}C_{10}$ (D) $a - 2c = \frac{2^{10}(1.3.5 \dots 19)}{10!}$

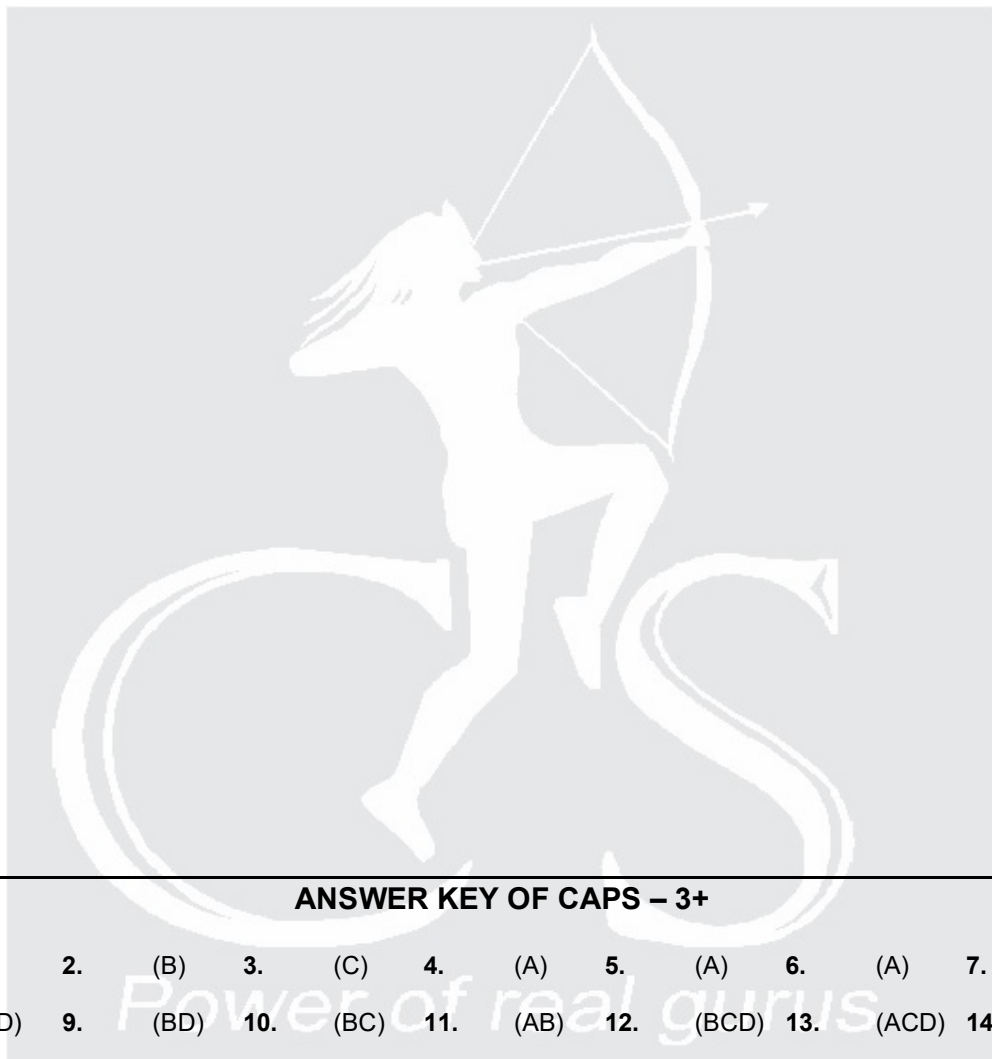
8. Let $(1+x^2)^2(1+x)^n = \sum_{k=0}^{n+4} a_k x^k$. If a_1, a_2, a_3 are in AP, then n is (given that ${}^nC_r = 0$, if $n < r$):
- (A) 6 (B) 4 (C) 3 (D) 2
9. If ${}^{100}C_6 + 4 \cdot {}^{100}C_7 + 6 \cdot {}^{100}C_8 + 4 \cdot {}^{100}C_9 + {}^{100}C_{10}$ has the value equal to xC_y ; then the possible value(s) of $x + y$ can be :
- (A) 112 (B) 114 (C) 196 (D) 198
10. For all real number x , let $f(x) = \frac{1}{2x-i} (i = \sqrt{-1})$. If there exists real numbers $a_1, a_2, a_3, a_4, a_5, a_6$ for which $f(a_1), f(a_2), f(a_3), f(a_4), f(a_5), f(a_6)$ form a regular hexagon on the complex plane, then which of the following is/are true :
- (A) Area of hexagon is $6\sqrt{3}$ sq. units
- (B) Area of hexagon is $\frac{3\sqrt{3}}{8}$ sq. units
- (C) Centre of circle circumscribing hexagon is $\left(0, \frac{1}{2}\right)$
- (D) Centre of circle circumscribing hexagon is $\left(0, -\frac{1}{2}\right)$
11. Let z_1, z_2, z_3 are three complex numbers such that $|z_1| = |z_2| = |z_3| = 1$ and $\frac{z_1^2}{z_2 z_3} + \frac{z_2^2}{z_1 z_3} + \frac{z_3^2}{z_1 z_2} + 1 = 0$ then integral values that $|z_1 + z_2 + z_3|$ can assume is
- (A) 1 (B) 2 (C) 3 (D) 4
12. Each of 2010 boxes in a line contains one red marble, and for $1 \leq k \leq 2010$, the box at the k th position also contains k white marbles. A child begins at the first box and successively draws a single marble at random from each box in order. He stops when he first draws a red marble. Let $P(n)$ be the probability that he stops after drawing exactly n marbles. The possible value(s) of n for which $P(n) < \frac{1}{2010}$, is
- (A) 44 (B) 45 (C) 46 (D) 47
13. A slip of paper is given to a person 'A' who marks it with either a plus or a minus sign. The probability of his writing plus is $\frac{1}{3}$. 'A' passes the slip to 'B' who may either leave it as it is or change the sign before passing it to 'C'. Similarly 'C' passes it on to 'D' and 'D' passes it to a referee who sees the plus sign on the slip. If 'B', 'C', 'D' change the sign with probability $\frac{2}{3}$. The probability that A originally wrote the plus sign is
- (A) less than $\frac{1}{3}$ (B) greater than $\frac{9}{13}$
- (C) less than $\frac{13}{25}$ (D) lies in interval $\left(\frac{1}{4}, \frac{1}{2}\right)$

Numerical Based Questions :

14. If ${}^{25}C_0 \cdot {}^{25}C_2 + 2 \cdot {}^{25}C_1 \cdot {}^{25}C_3 + 3 \cdot {}^{25}C_2 \cdot {}^{25}C_4 + \dots + 24 \cdot {}^{25}C_{23} \cdot {}^{25}C_{25} = k \cdot {}^{49}C_\lambda + {}^{50}C_\mu$, then find the value of $2k - \lambda - \mu$. (where $\mu, \lambda < 25$)
15. All the face cards from a pack of 52 playing cards are removed. From the remaining pack half of the cards are randomly removed without looking at them and then randomly drawn two cards simultaneously from the remaining. If the probability that two cards drawn are both aces is $\frac{P({}^{38}C_{20})}{{}^{40}C_{20} \cdot {}^{20}C_2}$, find P.
16. Number of five digits numbers divisible by 3 that can be formed using the digits 0, 1, 2, 3, 4, 7 and 8 if, each digit is to be used atmost one is N where 'N' is three digit number then sum of the digit of 'N'
17. In a shooting competition a man can score 0, 2 or 4 points for each shot. Then the number of different ways in which he can score 14 points in 5 shots, is N then number of digits in 'N' equals
18. The number of ways in which 5 X's can be placed in the squares of the figure so that no row remains empty is:

C				
B				
A				

19. Let $z_1, z_2 \in \mathbb{C}$ such that $|z_1 - 3i| + |z_1 + 4| = 5$ and $|z_2 - 3i| + |z_2 + 4| = 8$. If maximum value of $|z_1 - z_2|$ is $\frac{p}{q}$ where p, q are relatively prime numbers then find the value of $(p - 2q)$.
20. Find the number of ordered pairs (a, b) of real numbers such that $(a + ib)^{2015} = a - ib$
21. If $A_1, A_2, \dots, A_n$ be the vertices of an n-sided regular polygon such that $\frac{1}{A_1 A_2} = \frac{1}{A_1 A_3} + \frac{1}{A_1 A_4}$, then find the value of n
22. Rice University and Stanford University write questions and corresponding solutions for a high school Math tournament. The Rice group writes 10 questions every hour but make a mistake in calculating their solutions 10% of the time. The Stanford group writes 20 problems every hour and makes solution Mistakes 20% of the time. Each school works for 10 hours and then sends all problems to Smartie to be checked. However, Smartie isn't really so smart, and only 75% of the problems she thinks are Wrong are actually incorrect. Smartie thinks 20% of questions from Rice have incorrect solutions, and that 10% of questions from Stanford have incorrect solutions. This problem was definitely written and solved correctly. The probability that Smartie thinks its solution is wrong is $\frac{1}{n^2} (n \in \mathbb{N})$, then determine n.
23. There are 10 cards, labeled from 1 to 10. Three cards denoted by a, b, c ($a > b > c$) are drawn from the cards at the same time if the probability such that $\int_0^a (x^2 - 2bx + 3c)dx = 0$ is $\frac{1}{k}$ then $\frac{k}{10}$ is.



ANSWER KEY OF CAPS – 3+

1.	(D)	2.	(B)	3.	(C)	4.	(A)	5.	(A)	6.	(A)	7.	(BD)
8.	(BCD)	9.	(BD)	10.	(BC)	11.	(AB)	12.	(BCD)	13.	(ACD)	14.	5
15.	6	16.	15	17.	2	18.	98	19.	9	20.	2017	21.	7
22.	5	23.	4										